Practical No.: 5

Title: Creating a Simple Login Form in ReactJS

Aim:

To develop a **simple login form** in ReactJS that accepts a username and password and displays a success message upon correct input.

Theory

A **login form** is used to authenticate users based on their credentials. In **ReactJS**, forms are handled using **controlled components**, where input values are managed using **state**. The `useState` hook is used to track user input dynamically.

Concepts Covered:

- Controlled Components
- Handling form inputs (text and password fields)
- Form submission handling
- Simple validation

Syntax for Handling Input in ReactJS:

```
const [value, setValue] = React.useState("");

const handleChange = (event) => {
    setValue(event.target.value);
};
```

.ReactJS Code

Step 1: Create `index.html`

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Practical 5: Simple Login Form</title>
    <script src="https://unpkg.com/react@18/umd/react.development.js"></script>
    <script src="https://unpkg.com/react-dom@18/umd/react-
dom.development.js"></script>
    <script src="https://unpkg.com/babel-standalone@6/babel.min.js"></script>
</head>
<body>
    <h2>Practical 5: Simple Login Form</h2>
    <div id="root"></div>
    <script type="text/babel" src="demo.js"></script>
</body>
</html>
```

Step 2: Create a JavaScript file (app.js)

```
function LoginForm() {
    const [username, setUsername] = React.useState("");
    const [password, setPassword] = React.useState("");
    const [message, setMessage] = React.useState("");

    const handleSubmit = (event) => {
        event.preventDefault();

        // Simple validation check
        if (username === "admin" && password === "12345") {
            setMessage("Login Successful! Welcome, " + username);
        } else {
            setMessage("Invalid Credentials. Try again.");
        }
    };

    return (
        <div>
            <h3>Login Form</h3>
            <form onSubmit={handleSubmit}>
                <label>
                    Username:
                    <input
                        type="text"
                        value={username}
                        onChange={(e) => setUsername(e.target.value)}
                        required
                    />
                </label>
                <br /><br />

                <label>
                    Password:
                    <input
                        type="password"
                        value={password}
                        onChange={(e) => setPassword(e.target.value)}
                        required
                    />
                </label>
                <br /><br />

                <button type="submit">Login</button>
            </form>
            <br />
            <p>{message}</p>
        </div>
    );
}

const root = ReactDOM.createRoot(document.getElementById("root"));
root.render(<LoginForm />);
```